SUPPORTING ADULT GUIDANCE

Inequalities on a number line

Equations, Inequalities and Simultaneous Relationships · Lesson 10. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson introduces solving **linear inequalities** and representing their solutions on a **number line**. Solving mirrors equation work, with the single new rule that multiplying or dividing by a negative reverses the inequality. Representation makes the solution set explicit: open/closed endpoints and a direction. It also touches double (bounded) inequalities as a stretch.

Driving Question: *What if "equals" becomes "at least" or "at most"?*

The mathematics, in plain terms

A linear inequality is solved by the same inverse operations as the corresponding equation, preserving the inequality — except that multiplying or dividing both sides by a negative number reverses its direction. The solution is a half-line (a ray), drawn with an open endpoint for strict inequalities and a closed endpoint for non-strict ones. A double inequality bounds x on both sides, giving a segment.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — A two-step inequality

Solve $2x + 1 > 7$ and show it on a number line.

1. Subtract 1 from both sides (as for an equation).	$2x > 6$
2. Divide by 2 — positive, so no flip.	$x > 3$
3. Represent it: > means an open circle, pointing right.	open circle at 3, arrow right

$x > 3$ — all values greater than 3 (3 itself excluded, so an open circle).

Example 2 — When the sign flips

Solve $5 - 2x \geq 1$ — watch the flip.

1. Subtract 5 from both sides.	$-2x \geq -4$
2. Divide by –2 — negative, so REVERSE the sign.	$x \leq 2$
3. Represent it: ≤ means a closed circle, pointing left.	closed circle at 2, arrow left

$x \leq 2$ — dividing by –2 flipped ≥ to ≤; 2 is included, so a closed circle.

Solve like an equation — mind the flip

Reassure learners that solving is the familiar process. Show — do not assert — the one new idea: multiplying by a negative reflects both sides of a true inequality across 0 ($2 < 5$ becomes $-2 > -5$), so the sign must reverse to stay true. Let them see it before they use it. Make representation routine: after solving, always draw the number line and say the range in words. The two common errors are forgetting the flip and mixing up open/closed endpoints.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Read the four kinds of range on a number line (open/closed, direction).	~9 min
Movement	Solve inequalities — including the negative sign-flip.	~11 min
Reflection	Solve, then represent the answer on a number line.	~12 min
Creativity	Solve a double inequality and draw the bounded segment.	~8 min
Rest	A short pause on 'a range of acceptable answers'.	~2 min
Nutrition	Intellectual: the everyday logic of constraints — at least / at most.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Forgetting to flip when dividing/multiplying by a negative.
Test a value: 'Is $x = 0$ a solution of $-x < 3$? Yes. Does your answer include 0? If not, the sign should have flipped.'

Mixing up open and closed circles.
Tie the circle to the sign: 'Does the boundary count? ≤ and ≥ include it (closed ●); < and > exclude it (open ■).'

Drawing the arrow the wrong way.
Read the solved inequality: '$x > 3$ — bigger than 3, so which way along the line?' Point to larger values for > ≥.

Treating the answer as a single number.
'Is 4 a solution? Is 5? So the answer is not one value — what does the whole range look like?'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the number line and directed (negative) numbers, and solving a two-step equation. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner divides $-x < 3$ to get $x < -3$, or uses a closed circle for a strict inequality. The aim is a question that returns the reasoning.

- "Is $x = 0$ a solution of $-x < 3$? Does your answer include 0 — or did the sign need to flip?"

- “Does the boundary value itself count? So is the circle open or closed?”
- “x > 3 — which way does the arrow point, towards bigger or smaller values?”
- “Name two values in your range and one just outside it — do they behave as they should?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why does the sign flip for a negative?

Multiplying by a negative reverses order on the number line: 2 < 3, but –2 > –3. So to keep the statement true, the inequality must reverse. A quick numeric check convinces most learners.

Open or closed — how do they remember?

'Or equal to' (≤, ≥) includes the boundary → closed ●. Strict (<, >) excludes it → open ■. Reading the inequality aloud usually settles it.

Do boundaries have to be whole numbers?

No — the stretch gives $x \leq 2.5$. Place the circle between the integer ticks; the method is unchanged.

How does this connect to the next lesson?

A one-variable inequality is a range on a line; a two-variable inequality is a region in the plane. Lesson 11 extends open/closed boundaries and shading from the line to a graph.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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